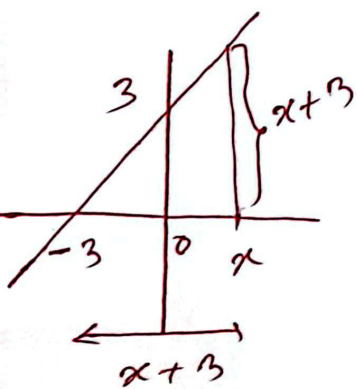


1. Sketch the area represented by $f(x) = \begin{cases} 5, & 0 \leq x \leq 2 \\ 3+x, & 2 < x \leq 4 \end{cases}$ for $\int_0^4 f(x) dx$. Evaluate the integral using Fundamental theorem of Calculus. [4]

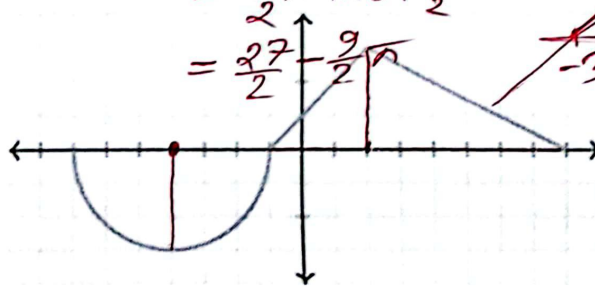
2. Graph the function $f(x) = x + 3$ over the interval $[-3, x]$. Then use simple area formula from geometry to find the area function $A(x)$ that gives the area under the graph and confirm that $A'(x) = f(x)$. [4]

3. According to following figure evaluate $\int_{-7}^8 f(x) dx$. [2]



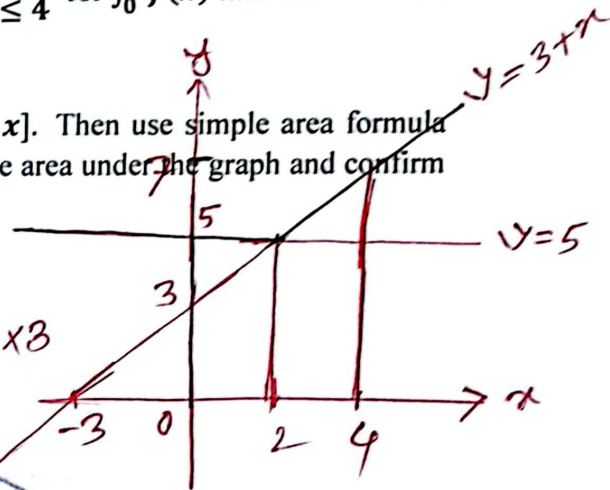
$$A(x) = \frac{1}{2} (x+3)(x+3) = \frac{1}{2} (x+3)^2$$

$$A'(x) = \frac{1}{2} \times 2(x+3) = x+3 = f(x)$$



$$= -\frac{1}{2} \times \pi \times 3^2 + \frac{1}{2} \times 3 \times 3$$

$$= \frac{27}{2} - \frac{9\pi}{2}$$



$$\int_0^4 f(x) dx$$

$$= \int_0^2 f(x) dx + \int_2^4 f(x) dx$$

$$= 2 \times 5 + \frac{1}{2} (5+7) \times 2$$

$$= 10 + 12 = 22$$